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ODD END SEMESTER EXAMINATION DECEMBER 2025

Name of the Course: B. Tech. Technology

Semester: V

Name of the Paper: Stem Cell Technology

Paper Code: TBT 508

Time: 3 Hours

Maximum Marks: 100

Note:-

1. All questions are compulsory
2. Answer any two sub questions among a, b, & c in each main question
3. Total marks in each main question are twenty
4. Each question carries 10 marks

Q1	(20 marks)	
(a)	Explain the different methods for fate mapping of stem cells.	CO1
(b)	Define stem cells and state the different types of stem cells based on their characteristics.	
(c)	Describe the milestones of advances in stem cell biology.	
Q2	(20 marks)	
(a)	Discuss the different types of adult stem cells, transdifferentiation and its applications.	CO2
(b)	Explain induced pluripotent stem cells.	
(c)	Briefly describe the potency assays for measuring engraftment potential of hematopoietic and progenitor cells.	
Q3	(20 marks)	
(a)	Elaborate on the different modes of cell signalling along with the various kinds of receptors.	CO3
(b)	Define cell commitment and differentiation. Describe the factors affecting cell differentiation.	
(c)	Discuss the various stages of embryogenesis.	
Q4	(20 marks)	
(a)	Describe embryonic stem cells with emphasis on its production and maintenance.	CO4
(b)	What are stem cell markers. Discuss the different types of stem cell markers.	
(c)	Explain the process of therapeutic cloning with stem cells.	
Q5	(20 marks)	
(a)	Explain tissue engineering and its role in regenerative medicine.	CO5
(b)	Discuss the role of stem cells in the treatment of diabetes.	
(c)	Describe hematopoietic stem cell transplantation (HSCT).	

Note for the question paper setters:

